Case 03 · Architecture decision log (ADR): Roll Transparency Log

Every decision that could have gone the other way, in ADR form: context, options, decision, price, and what would have happened otherwise.

ADR-01: Who runs the log?

Status: accepted. **Context:** only the Commission can write to the roll. **Options:** an independent body, or the Commission with witnesses. **Decision:** the Commission, with witnesses; trust comes from who approves. **Price:** dependence on its availability.

Otherwise: an independent body received weekly files and the log was always behind.

ADR-02: Plain hash or keyed identifier?

Context: elector numbers are guessable. **Decision:** HMAC with a key inside an HSM. **Price:** an HSM to operate.

Otherwise: anyone could build a dictionary of every number.

ADR-03: When is a change logged?

Decision: before it takes effect; no "saved" without a receipt. **Price:** the log is a dependency of every change.

Otherwise: changes could skip the log and be noticed only at publication.

ADR-04: How many witness signatures?

Decision: three of five. **Price:** coordination with outside bodies.

Otherwise (one): a single compromised witness enables a split view.

ADR-05: What does everyone see?

Context: the POC found almost every booth has fewer than 5 changes. **Decision:** aggregates by district and week. **Price:** less detail for researchers.

Otherwise: the public log is a map of who left the village.

ADR-06: Keep an "other" code?

Decision: no; 11 codes. **Price:** officers must choose.

Otherwise: "other" becomes the new opaque bucket.

ADR-07: Tessera or Trillian?

Decision: Tessera, static tiles on a CDN. **Price:** a newer project.

Otherwise: a live read server must survive draft-day load.

ADR-08: Reconcile daily or only at publication?

Decision: both. **Price:** a nightly job per booth.

Otherwise: a bypassing change is found weeks later.